2 STAGE OP-AMP

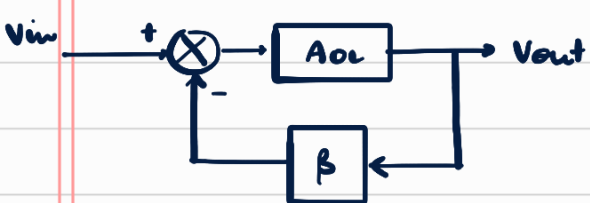
STAGE - I : 5T OTA
II : CS amp

- single stage op-amp (OTA)
- WHY 2 STAGE ?
- miller cap - ?

: KEY benefits of 2stage opamp

1) HIGH GAIN

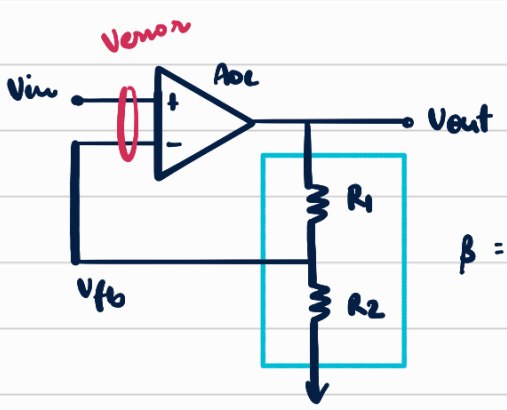
* Why need high gain Amp ?



$$\frac{V_{out}}{V_{in}} = \frac{A_{OL}}{1+\beta \cdot A_{OL}} \approx \frac{1}{\beta} = A_{CL}$$

$A_{OL} \gg \beta$ ∴

we need this high gain to put the amp in neg. feedback



$$\beta = \frac{R_2}{R_1+R_2} = \frac{V_{fb}}{V_{out}}$$

$$\therefore A_{CL} = \frac{R_1+R_2}{R_2} = 1+\frac{R_1}{R_2} \;//$$

* high gain :
Vout ↑↑ or
Vin ↓↓
as the Verror gets smaller,
the $A_{CL} \approx \frac{1}{\beta}$, i.e
amplifier's
behavior is now dictated
by its precise passive resistors
instead of unreliable A_{OL}

A_{OL} : unreliable
— varies with temp, voltage,
frequency, process variation.

HIGH gain enables good neg. feedback.

high gain is needed so that neg. feedback can work precisely and reduce the unstable effects of A_{OL} on A_{CL}

* feedback EXPANDS BW by $1 + \beta \cdot A_{OL}$

the GBP remains the same

$A_{OL} \cdot \omega_{p\,OL} = A_{CL} \cdot \omega_{p\,CL}$ $\quad \omega_p$ = BW = −3dB freq

$\omega_{p\,CL} = (1 + \beta \cdot A_{OL})\, \omega_{p\,OL}$

$UGB_{Thms.} = \omega_p \cdot A$ → where gain = 0 dB

On unity gain FB → $A_{CL} = 1$ i.e $\beta = 1$

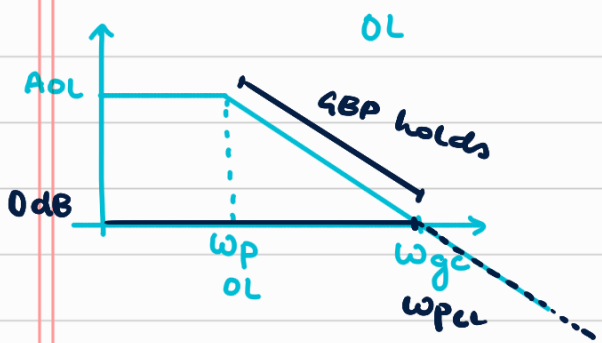
gain cross freq = CL BW = ω_{gc} → −3dB pole of CL

where A_{OL} curve goes to 0dB

max fb ⟶ min gain ⟶ max BW

only when $A_{OL} \gg 1$

$\because \omega_{p_{CL}} = (1 + A_{OL})\, \omega_{pOL}$



* **RISK OF INSTABILITY :**

Barkhausen's oscillation criteria :

$\beta \cdot A_{OL} = 1 \angle -180°$ ⟶ makes it an oscillator

$\therefore$ All other poles are made such they come after the UGB freq. so that phase doesn't reach $-180°$



$$\beta A_{OL}(s) = \frac{A_0 (1 + s/\omega_z)}{(1 + \frac{s}{\omega_{p_1}})(1 + \frac{s}{\omega_{p_2}})}$$

* SMALL PM ⟶

in the CL : peakings ⟶ noise amplification (freq domain)
ringings ⟶ poor settling time, overshoot
(time domain)
(Step response)

∴ freq compensation procedure

$\omega_{gc} = \beta \cdot UGB$ (GBP)

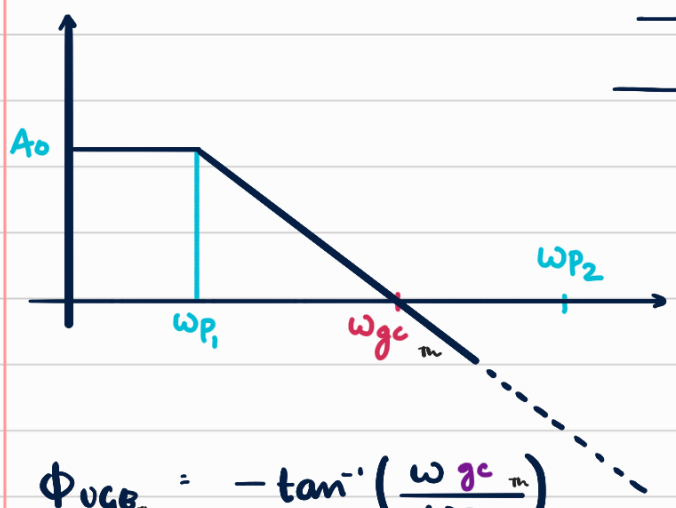
control PM

control : $\frac{\omega_{p_2}}{\omega_{gc\ Theo.}}$

→ ω_{p_1} already introduces a -90° ϕ

→ ω_{p_2} also adds another -90°

→ closer ω_{p_2} is to ω_{gc} Theoretical, more closer the total ϕ is to -180°



$\phi_{UGB_{Theo.}} = -\tan^{-1}\left(\frac{\omega_{gc\ Th}}{\omega_{p_1}}\right)$

from single pole alone:

$A_{OL} \cdot \omega_{p_1} = 1 \cdot \omega_{gc}$

$\frac{\omega_{gc\ Th}}{\omega_{p_1}} = A_{OL}$

$\therefore \phi_{gc\ Th.} \approx -90^\circ$

open LG

for $A_{OL}(s)$
0dB ⟶ UGB
for $\beta \cdot A_{OL}(s)$ LG
0dB ⟶ ω_{gc}

* PM & GM calc. in LG

* $\omega_{gc} = \beta \cdot UGB$

if $\omega_{p_2} = \omega_{gc\ theoretical}$

theoretically $\phi_{TOT} = -90^\circ + (-45^\circ)$

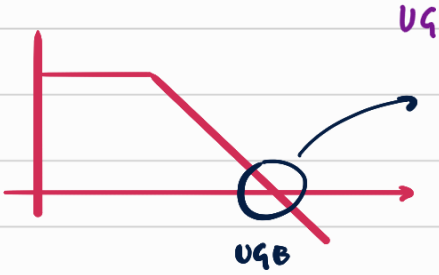
$\therefore$ PM = 45°

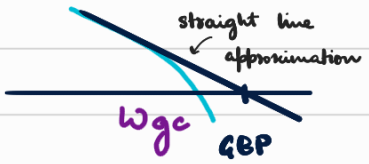
BUT PM actually > 45°

this is because the 2nd pole causes some bending below the asymptote

$\therefore \omega_{gc} < \underbrace{\text{Gain BW product}}_{UGB}$ | $A_{OL} = 1$

since ω_{gc} is actually smaller than the theoretical value, $\therefore \phi_{gc}$ introduces is less than 45°





$\therefore \phi_{TOTAL} = -90^\circ + (-\phi^\circ)$

$\therefore$ **PM > 45°**

as ω_{p_2} is closer to GBP, the gc actual starts deviating more.

at $\frac{\omega_{p_2}}{gc} \longrightarrow \infty$, $UGB \approx UGB_{actual}$

* UGB theoretical = GBW

OPTIMAL PM :

1. TRANSIENT response ⟶ CL : fast settling, no overshoot ↓

$\therefore$ critically damped $\quad \therefore \zeta = 1, \; Q = \frac{1}{2\zeta} = 0.5$

$\Longrightarrow \frac{\omega_{p_2}}{\omega_{gc}} = 4 \longrightarrow$ PM = 76°

$$A_{OL}(s) = \frac{A_0}{(1+\frac{s}{\omega_{p_1}})(1+s/\omega_{p_2})}$$

characteristic equation : $1 + \beta \cdot A_{OL}(s)$

$$(1+\frac{s}{\omega_{p_1}})(1+\frac{s}{\omega_{p_2}}) + \beta \cdot A_0 = 0$$

$$\Rightarrow \quad 1 + s\left(\frac{1}{\omega_{p_1}} + \frac{1}{\omega_{p_2}}\right) + \frac{s^2}{\omega_{p_1} \cdot \omega_{p_2}} + \beta A_0 = 0$$

$$\frac{s^2}{\omega_{p_1} \cdot \omega_{p_2} \cdot (1+A_0\beta)} + s\left[\frac{\omega_{p_1} + \omega_{p_2}}{\omega_{p_1} \cdot \omega_{p_2} \cdot (1+\beta A_0)}\right] + 1 = 0$$

compare to $\frac{s^2}{\omega_n^2} + \frac{2\zeta}{\omega_n} \cdot s + 1 = 0$

$$\omega_n = \sqrt{\omega_{p_1} \cdot \omega_{p_2} \cdot (1+\beta A_0)}$$

$$\frac{2\zeta}{\omega_n} = \frac{\omega_{p_1} + \omega_{p_2}}{\omega_n^2}$$

$$\zeta = \frac{\omega_{p1} + \omega_{p2}}{2\omega_n} = 1$$

$$= \frac{\sqrt{\omega_{p1}}}{\sqrt{\omega_{p1} \cdot \omega_{p2} \cdot (1+\beta A_0)}} + \frac{\sqrt{\omega_{p2}}}{\sqrt{\omega_{p1} \cdot \omega_{p2} \cdot (1+\beta A_0)}} = 2$$

now $\omega_{p2} \gg \omega_{p1}$

$$\therefore \frac{\omega_{p2}}{\omega_{p1}(1+\beta A_0)} = 4$$

now $A_{OL} \cdot \omega_{p1} = A_{CL} \cdot \omega_{pCL} = UGB$, $A_{CL} = \frac{1}{\beta}$

$\omega_{pCL} = (1 + A_0\beta)\omega_{p1} \approx \beta \cdot A_0 \omega_{p1}$

$\therefore \beta A_0 \cdot \omega_{p1} = \omega_{gc}$

$\therefore \omega_{pCL} = \beta \cdot UGB$

$= \omega_{gc}$

when $A_{OL} \gg$ high

$$\therefore \frac{\omega_{p2}}{\omega_{gc}} = 4$$

ii) freq response (CL)

→ no peaking, maximally flat at $\zeta = 0.707$ (underdamped tho)

$$\therefore \frac{\omega_{p2}}{\omega_{gc}} = 2 \quad PM = 66^\circ$$

4 * no ringing, overshoots → PM ≥ 76°

2 * no peaking and MAX BW_{CL} PM ≈ 66° (40% higher BW compared to 1st order)

✓ * faster response PM ≈ 72°

$\frac{\omega_{p2}}{\omega_{gc}} = 3$

2nd pole → makes the system faster

→ increases BW

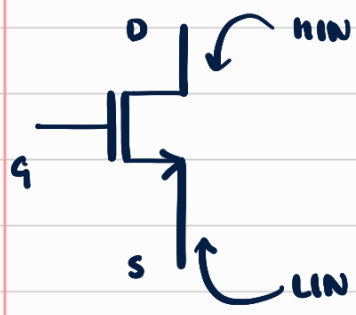
Why 2 Stage Amplifier ?

* each node in a circuit is a POLE !

$$\omega_p = \frac{1}{R_{in} \cdot C}$$

* HINs (high impedance nodes) MOSFETS gives dominant poles

$$A_V = g_m \cdot R_{in}$$



* LINs (low impedance nodes) now dominant poles

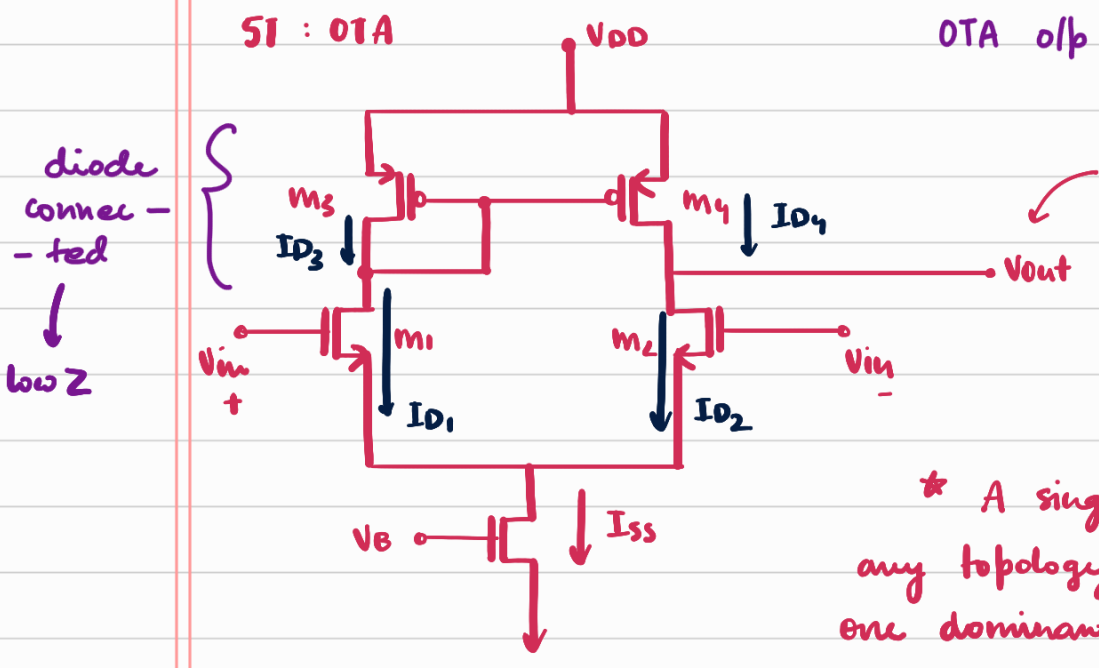
we want a single high impedance node, and all others far away for stability

* provided by a single stage amp

voltage in, current out
HIGH o/p impedance

5T : OTA

OTA o/p



diode connected → low Z

$R_{in} = r_{o2} \parallel r_{o4}$ high o/p impedance

Differential amp i/p

* A single stage amp is any topology that produces one dominant - HIN o/p with no second independent amplifying feeding from it.

$r_{o\,eq} = \frac{r_o}{2}$

$A_V = g_m \cdot r_{o\,eq}$

$= \frac{2I_D}{V_{gs} - V_{th}} \cdot \frac{r_o}{2}$

$= \frac{I_D}{V_{gs} - V_{th}} \cdot \frac{V_A}{I_D}$

$$\boxed{A_V = \frac{V_A}{V_{gs} - V_{th}}}$$

V_{out} at drain of both M2 & M4 $\therefore$ R_{out} = parallel of r_{o4} & r_{o2} assuming $r_{o2} \approx r_{o4} \approx r_o$

$\therefore$ gain is limited by device parameters

* V_A : how much I_D changes with V_{DS}

$V_A \propto L$

We want high gain in order for our negative feedback to work precisely, so that it can eliminate the unstable effects of Aol from coming into the Acl, for stabilty a single HIN is best because it provides a phase lag of 90deg at like infinity meaning atleast 90 deg PM, **unconditionally** stable. But we know that a single stage can't provide us a high enough gain (because of being limited by device parameters) for the negative feedback to work properly, so we use 2 stage amplifier.

use a cascode amp :
1. Telescopic cascode
2. Folded cascode

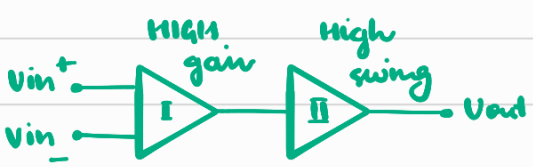
} not very good for low supply voltage → limited headroom (signal swing)

= high gain $(g_m r_o)^2$

2 STAGE AMPLIFIER

isolates our high gain & signal swing requirements.

STAGE I ⟶ ST OTA or cascode
II ⟶ common source



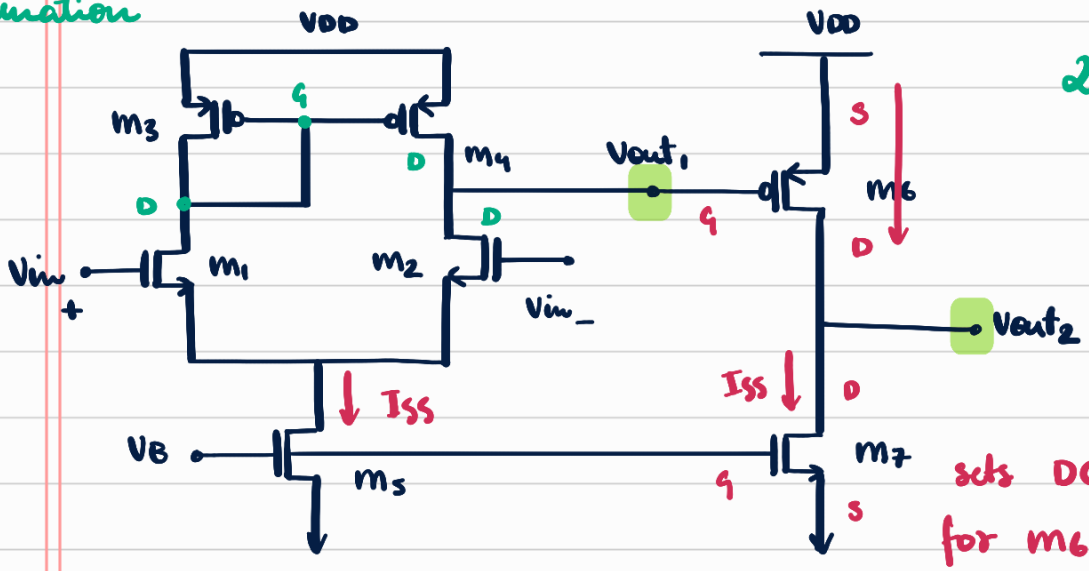
* 2 competing choices

1. HIGH gain ⟶ wants voltage swings to be less
2. LARGE signal swing ⟶ so that V_{ov} doesn't go over V_{DS} (amp. needs saturation region)

⟶ to use as much of the voltage range as possible

* for better SNR too

* stage II has more swing room even tho m_6, m_7 are also required to be in saturation



2 HINs

sets DC operating for m_6

* m_6 source ⟶ V_{DD}

change in Vout 1 ∴ V_{gs} changes

PMOS, : if Vout 1 rises, V_{gs} becomes less −ve ∴ conducts less current

m_6 drain • m_7 drain ⟶ $Vout_2$

m_7 as a current source

∴ m_6 gives LESS current

m_7 wants Iss ∴ parasitic caps at $Vout_2$ supply the rest and $Vout_2$ FALLS

∴ Vout 1 rises, Vout2 falls

* HIGH IMP. nodes ⟶ 2 (can reduce STABILITY)

∴ 2 dominant poles

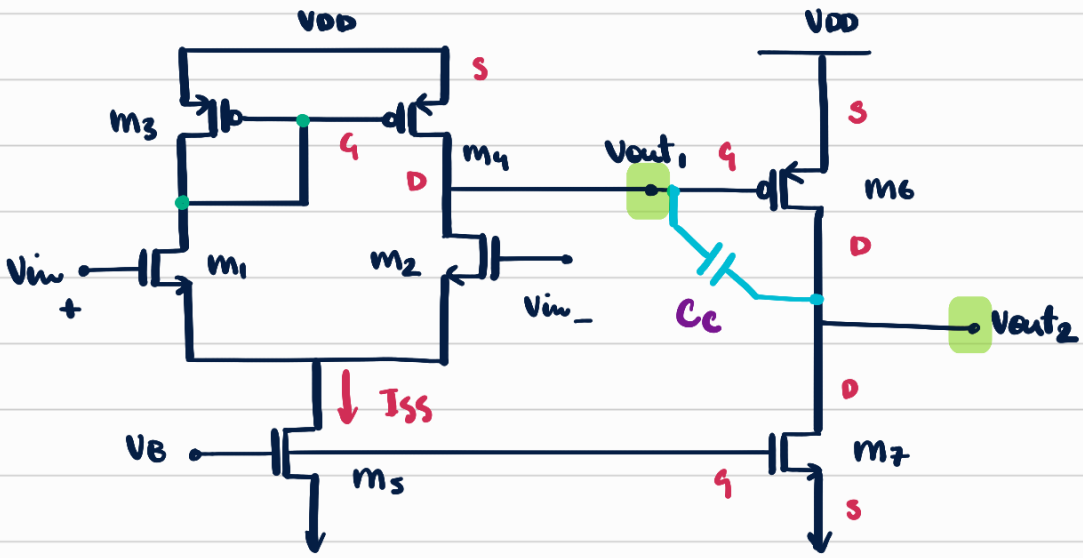
internal node

$V_{out_1} \Rightarrow \omega_{p_1} = \frac{1}{R_{out_1} \cdot C_1}$

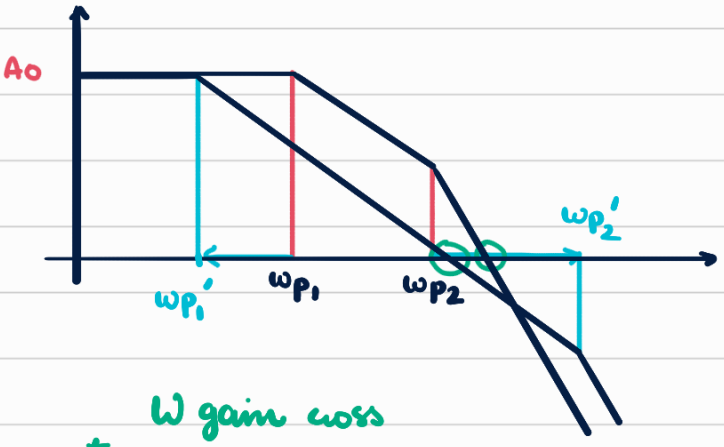
output node

$V_{out_2} \Rightarrow \omega_{p_2} = \frac{1}{R_{out_2} \cdot C_2}$

} use miller capacitance COMPENSATION : SPLITS THE POLES!



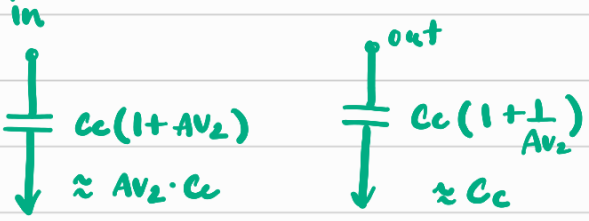
* pushes ω_{p_1} inwards
* pushes ω_{p_2} outwards



* W gain cross shifts inside



A_{v_2}: -ve inverting amp



* creates a large enough pole separation for dominant pole approximation to hold true!

$\therefore$ equivalent Cap at $V_{out_1} = (C_1 + C_c \cdot A_{V_2})$

since $C_c \gg C_1$ and A_{V_2} is large $\therefore \approx C_c \cdot A_{V_2} = g_{m_2} \cdot R_{out_2} \cdot C_c$

since by approximation ω_{p_2} is at a high enough freq. here we can assume C_c to SHORTED.

$\therefore$ M_6 becomes diode connected

$\therefore R_{out_2} = \frac{1}{g_{m_2}}$ (very less)

$C_{eq} = (C_1 + C_2)$ (parallel)

$\therefore$

$$\omega_{p_1} = \frac{1}{R_{out_1} C_1} \Longrightarrow \frac{1}{R_{out_1}(g_{m_2} \cdot R_{out_2} \cdot C_c)}$$

$$\omega_{p_2} = \frac{1}{R_{out_2} \cdot C_2} \Longrightarrow \frac{g_{m_2}}{(C_1 + C_2)}$$

POLES SPILT SUCCESSFULLY

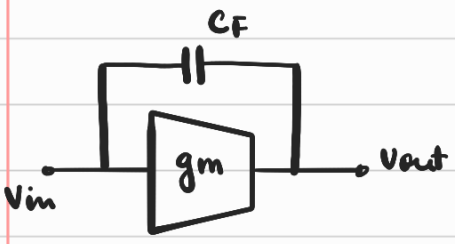
$\Longrightarrow$ What miller did ?

a.) increased the effected C_{in} by A_{V_2} to push ω_{p_1} even lower.

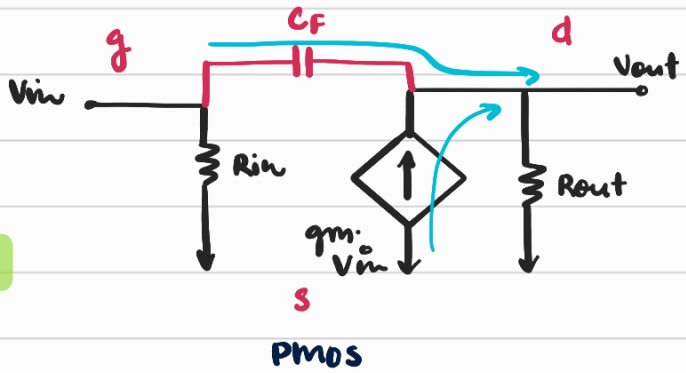
b.) makes V_{out_2} a LIN at high freq $\therefore$ pushing ω_{p_2} higher

* effectively making it a single HIN circuit !!

* BUTTT..... an undesirable effect of a feed forward ZERO.



any cap bridging the i/p and o/p of a gain creates signal feed forward path.



and thus a zero.

To find this zero : $V_{out} = 0$

$\therefore (V_{in} - \cancel{V_{out}}^{0}) sC_F + g_m \cdot V_{in} = \frac{\cancel{V_{out}}^{0}}{R_{out}}$

$V_{in}(sC_F + g_m) = 0$

$s = -\frac{g_m}{C_F}$ // LH zero

if g_m +ve

* RH plane
→ poles $+\phi^\circ$
zeros $-\phi^\circ$

* LH plane
→ poles $-\phi^\circ$
zeros $+\phi^\circ$

in CS (inverting amp)

A_V = -ve = $g_m r_o$

$\therefore g_m$ -ve

zero : $\frac{g_m}{C_F}$ in RHP

$\therefore \omega_z = \frac{g_{m2}}{C_{gd} + C_c} \approx \frac{g_{m2}}{C_c}$

RHP zero ——— bad for both mag. & phase

gain : +20dB/dec

phase : -90°

introducing a RHP zero

pushes PX inwards → reduces GM

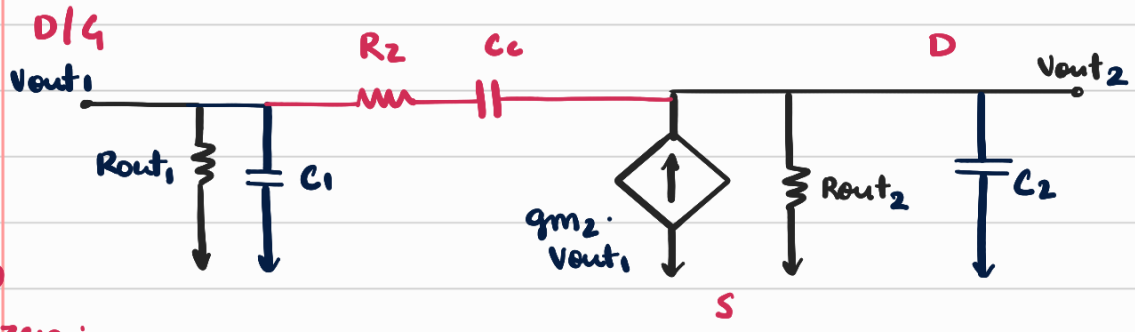
GX outwards → reduces PM

increasing C_c → for more pole-splitting benefit

→ decreases ω_z (RHP zero) closer to the gain crossover region, hurting the PM sooner

TO HANDLE : RHP ZERO

Resistance in series



$V_{out2} = 0$

to find zero :

$$\frac{(V_{out1} - \cancel{V_{out2}})^{0}}{\left(R_2 + \frac{1}{sC_c}\right)} + g_{m2} \cdot V_{out1} = \frac{\cancel{V_{out2}}^{0}}{\frac{R_{out2} \cdot 1/sC_2}{R_{out2} + \frac{1}{sC_2}}}$$

$$\frac{\cancel{V_{out1}} \cdot sC_c}{1 + sR_2 \cdot C_c} = -g_{m2}\cancel{V_{out1}}$$

$$s \cdot C_c = -g_{m2} - sR_2 \cdot g_{m2} \cdot C_c$$

$$sC_c(1 + g_{m2} \cdot R_2) = -g_{m2}$$

$$s = \frac{-g_{m2}}{C_c(1 + g_{m2}R_2)}$$

g_{m2} is -ve

$$\omega_z = \frac{1}{C_c\left(\frac{1}{g_{m2}} - R_2\right)}$$

place this zero at ∞

$\therefore R_2 = \frac{1}{g_{m2}}$ | we can achieve this by using a diode connected M6 ☹ ✗ WRONG for this there has to be DC current flowing through the MOS. BUT C_c in series blocks it !!

using a PMOS in triode region (behaves like a Resistance)

Design Constraints

$$\omega_{p1} = \frac{1}{R_{o1}(g_{m2}R_{o2})\cdot C_c}$$

$$\omega_{p2} = \frac{g_{m2}}{C_1 + C_2} \approx \frac{g_{m2}}{C_L}$$

increasing C_c, pushes ω_{p1} inwards and eventually ω_{p2} saturates

BUT it decreases the GAIN BW PRODUCT $\therefore \omega_{gc}$

hence reducing the speed of the circuit.

$$C_1 < C_c < C_L$$

$C_L \hookrightarrow C_{LOAD}$

* $C_c \approx (0.3 \longrightarrow 0.5)C_L$

$$GBP = A_{v1}\cdot A_{v2}\cdot \omega_{p1}$$

$$= g_{m1}\cdot \cancel{R_{o1}} \cdot \cancel{g_{m2}\cdot R_{o2}} \cdot \frac{1}{\cancel{R_{o1}}\,\cancel{g_{m2}R_{o2}}\cdot C_c}$$

$$GBP = \frac{g_{m1}}{C_c} = \omega_{gc}$$

critically damped $\zeta = 1$

$Q = 0.5$ PM = 76°

$$\frac{\omega_{p2}}{\omega_{gc}} = 4 \longrightarrow \frac{g_{m2}}{C_L}\frac{C_c}{g_{m1}} = 4$$

$$\frac{g_{m2}}{g_{m1}} = \frac{4\cdot C_L}{C_c}$$

assume

$C_L = 2pF$

$\therefore C_c = 0.5\,C_L = 1pF$

$$\therefore \frac{g_{m2}}{g_{m1}} = 4(2) = 8$$

if both stages use the same g_m/I_D level (i.e the st 1 & II i/p devices operate at the same point on the efficiency curve

$$\therefore \frac{g_{m2}}{g_{m1}} = \frac{I_{D2}}{I_{D1}}$$

I_{D2} → drain current of M_6 (stage 2 i/p)

I_{D1} → drain current of M_1 or M_2 (stage 1 i/p)

$I_{B_1} = 2\,I_{D_1}$, $I_{B_2} = I_{D_2}$

(M_5) (M_7)

$\therefore \quad \frac{I_{B_2}}{I_{B_1}} = \frac{I_{D_2}}{2\,I_{D_1}} = (4)$

energy inefficient

∵ stage 2 needs a lot of current to get the high g_{m_2} ⟶ stability & good PM !!

80% power consumed in STAGE II

✓ still robust, works at low supply, and straight forward to design

* GBP target = 5MHz

$C_L = 2pF$, $C_c = 1pF$

nmos : $V_{TO} = 0.7$, $k_n = 110\mu$, $\lambda = 0.05$

pmos : $V_{TO} = -0.7$, $k_p = 50\mu$, $\lambda = 0.06$

both stages : g_m/I_D same ✓

$V_{OV} = V_{GS} - V_{TO} =$ 0.2 assume $\therefore$ $g_m/I_D = 10\ V^{-1}$

now $GBP = \frac{g_{m_1}}{C_c \cdot 2\pi}$

$\therefore$ $g_{m_1} = 31.42\mu\ A/V$

$\therefore$ $\frac{g_{m_2}}{g_{m_1}} = 8$, $g_{m_2} = 251.33\mu\ A/V$

now $R_2 = \frac{1}{g_{m_2}} = 3978.88\ \Omega \approx 4K\Omega$

* currents $I_{D_1} = \frac{g_{m_1} \cdot V_{OV}}{2} = 3.142\,\mu A$

$\therefore$ $I_{B1} = 6.28\,\mu A$

now $I_{B2} = 4I_{B1} = 25.13\,\mu A$

TRANSISTOR SIZING :

$$I_D = \frac{k_p}{2} \cdot \frac{W}{L} (V_{ov})^2$$

taking $L = 1\mu m$ for all

nmos M1, M2 : $W/L = 1.428$, $W = 1.43\,\mu m$

pmos M3, M4 : $W/L = 3.142$, $W = 3.14\,\mu m$

M5 : $W/L = 2.9$, $W = 2.9\,\mu m$

M6 : $W/L = 25.1$, $W = 25.1\,\mu m$

M7 : $W/L = 11.4$, $W = 11.4\,\mu m$

* calculations

$$A_{V1} = g_{m1} \cdot (r_{o2} \| r_{o4})$$
$$= 31.42\mu\,(2.9\,M)$$
$$\approx 91 = 39.19\,dB$$

$$A_{V2} = g_{m2} \cdot R_{out2}$$
$$\approx 90.98 = 39.17\,dB$$

$$r_{o2} = \frac{1}{\lambda_n \cdot I_{D2}} \approx 6.37\,M\Omega$$

$$r_{o4} = \frac{1}{\lambda_p \cdot I_{D4}} \approx 5.31\,M\Omega$$

$R_{out1} = r_{o2} \| r_{o4} \approx 2.9\,M\Omega$

$r_{o6} = 664\,K\Omega$

$r_{o7} = 797\,K\Omega$

$R_{out2} = 362\,K\Omega$

TOTAL DC gain = 78.37 dB

$$\omega_{p1} = \frac{1}{2\pi\, R_{out1} (g_{m2} R_{out2}) C_c}$$

$= 603.22\,Hz$

$$\omega_{p2} = \frac{g_{m2}}{2\pi C_L} = 20MHz$$

$W_2 = 47GHz$ far off enough

* $V_{bias} = V_{TH} + V_{OV}$

$= 0.7 + 0.2$

$V_{bias} = 0.9V$

* V_{in+} / V_{in-}

M5, $V_{OV} = 0.2 = V_{gs} - V_{th}$

sat : $V_{DS} \geq V_{OV}$

$\therefore V_{DS} \geq 0.2V$

M1, $V_s = 0.2$

$V_g = V_{in+}$

$V_{ov_1} = 0.2 = V_{gs} - V_{th}$

$0.2 + 0.7 = V_{in} - 0.2$

$V_{in+} = 1.1V$ lower bound ∵ any lower ⇒ M5 ⟶ triode region

M3, $V_g = V_d$ $V_{OV} = 0.2$

$V_s = V_{DD}$ $V_{sg} - |V_{th}| = 0.2$

$V_{sg} = 0.9V$

$V_s - V_d = 0.9$

$V_{DD} - 0.9 = V_d$

M1, $V_{DS} \geq V_{ov_1}$

$V_{DD} - 0.9 - 0.2 = V_{gs} - V_{th}$

$V_{DD} - 0.9 - 0.2 = V_{in} - 0.2 - 0.7$

$V_{in} \leq V_{DD} - 0.2$ upper bound

taking $V_{DD} = 3.3V$

$$1.1 \leq V_{in\pm} \leq 3.1$$

$$\therefore \quad V_{in+} = V_{in-} = \underline{\underline{2.1\,V}}$$

* currents

	theor.	prac.	% error
I_{B_1}	6.28 μA	6.76 μA	7.6
I_{D_1}	3.14 μA	3.38 μA	7.6
I_{D_2}	3.14 μA	3.38 μA	7.6
I_{B_2}	25.13μ	27.611 μA	9.8

TRANSIENT :

slew rate

$$I = C \cdot \frac{dv}{dt}$$

$$\frac{dv_{out_1}}{dt} = \frac{I_{B_1}}{C_c} = SR$$

$$\therefore \quad SR = \frac{6.28 \times 10^{-6}}{1 \times 10^{-12} \times 6}$$

$$SR = 6.28\,V/\mu s$$

actual : 6.425 V/μs

% error = +2.3%